

LLM-Guided Synthesis of Constraint-Adaptive Heuristics for Satellite–Ground Scheduling

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Abstract

Graph optimization heuristics can achieve strong aggregate performance while relying on decision rules that fail under changes in resource constraints. Effective adaptation requires identifying not only when action preferences should change, but also whether the input representation contains the information needed for that change. We propose an LLM-guided synthesis framework for constraint-adaptive satellite–ground scheduling that uses certified decision evidence to jointly refine graph representations and ranking rules. Its central mechanism detects ranking requirements that become contradictory under the current feature representation and directs synthesis toward the missing structural distinctions, rather than repeatedly adjusting an insufficient scoring interface. Two complementary mechanisms support this refinement. First, controlled offline interventions on resource-constraint configurations generate paired static scheduling problems in which the same competing actions remain individually feasible. Sound bounds on conditional completion values certify preference reversals or preservation under explicit boundary commitments, yielding relational specifications of when rankings should change or remain stable. Second, structural witnesses guide the LLM to compose distinguishing features from a typed graph-operation library and synthesize corresponding ranking rules. Feature–rule pairs are selected jointly for declared-interface acyclicity, complete-schedule quality, and feature-computation cost, making representation expansion targeted and cost-aware. We formulate contact selection as maximum-weight independent set optimization and study adaptation using a frozen program on held-out scheduling instances and constraint configurations. The synthesized program responds to the current conflict graph within a fixed feasibility-enforcing kernel, without online LLM or oracle calls. By linking representation refinement to certified decision conflicts, the framework addresses information limitations beyond rule-only repair while explicitly accounting for the computational cost of constraint adaptation.

CCS Concepts

• **Computing methodologies** → **Planning and scheduling**; • **Mathematics of computing** → *Combinatorial optimization*.

Keywords

Satellite scheduling, maximum-weight independent set, heuristic synthesis, representation refinement, large language models

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1 Introduction

A satellite–ground contact commits scarce resources and excludes competing opportunities. Their forgone reward depends on whether they could coexist. Resource thresholds change this relationship while contact windows and rewards remain fixed. Operational scheduling models coupled resources [2, 16]; we study pairwise-conflict scheduling, where a reusable maximum-weight independent set (MWIS) heuristic must respond to different static configurations.

LLMs synthesize executable heuristics using objective feedback in FunSearch, EoH and ReEvo [19, 21, 24]. DRAGON uses decomposition/reconstruction agents with boundary constraints and checker feedback [5]. We ask how synthesis can identify missing inputs that make required rankings impossible: formula revision cannot satisfy incompatible strict preferences over identical observations.

Figure 1 constructs this obstruction. Actions a, b have equal rewards and base features. Their forced-completion increments are 20, 26 before a ground-gap intervention and 20, 14 afterward. Both remain feasible; the preference reverses, but every pointwise base scorer ties them within each configuration. Shared resource fields cannot distinguish them; neighborhood structure can. These values concern forced completions, not an unrestricted rollout’s reward.

We propose certified representation–rule refinement. Following abstraction refinement and planning-feature precedents [6, 10], we project sound strict preferences onto exact feature-equivalence classes. Directed cycles certify an information obstruction. Witnesses retain ranking arcs and equality joins, targeting LLM proposals toward missing structural distinctions. Full refined quotients are repeatedly checked: separating one witness can leave other cycles intact.

Shared residual components cancel from value differences without bounding their unknown optima. Reached argmax audits retain finite-pool regret bounds and unknowns. Demand-driven execution shares typed computations and charges work. TRAIN selection balances complete-schedule reward and cost, with reached regret ranking within a declared utility band. Our contributions are:

- (1) **Certified information diagnosis**: finite pointwise representability characterized by quotients and concrete cycle witnesses.
- (2) **Witness-guided repair**: full-quotient separation and minimum additive-cost refinement over fixed evidence and a bounded finite catalogue.



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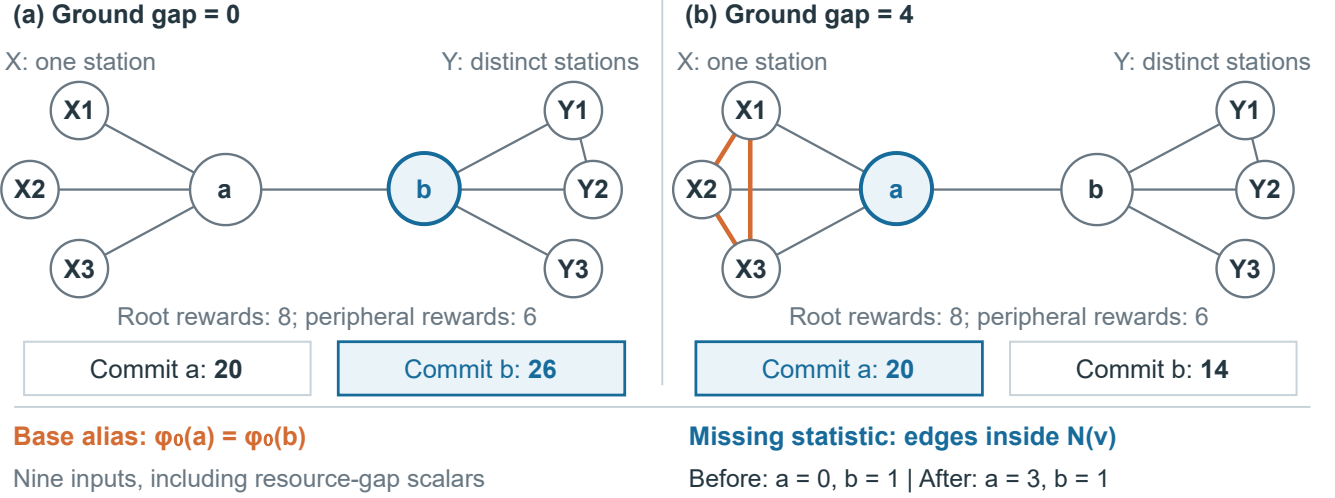


Figure 1: A ground-gap intervention reverses forced-completion preferences while preserving the nine-input base alias within each configuration. Neighborhood structure supplies the missing distinction. Values exclude the same fixed boundary reward.

- (3) **Decision-aware synthesis:** cancelled evidence, reached-action audits and demanded computation support joint selection.

One offline assistant session supplies 12 new proposals; existing controls have unequal generation counts and compute. TRAIN-only selection freezes programs before fresh scheduling and public-graph confirmation against published MWIS implementations. A fixed kernel enforces graph feasibility without online LLM or oracle calls; quality and transfer remain empirical.

2 Related Work

Scheduling and graph decisions. Mixed-integer decomposition [2], integrated collection/downlink planning [16], and ground-queue-aware traffic scheduling [22] establish resource context’s importance. Our pairwise model excludes arbitrary cumulative constraints. Graph embeddings condition optimization on partial solutions [7]; learned deferral changes independent-set decision stages [1]. AAMAS work combines factor-graph inference with routing [17] and local topology with neural representations [3]. Structural inputs are established choices. Published weighted branch-and-reduce [15], Struction [11], memetic kernelization [13] and concurrent local search [14] provide our specialized comparators.

GNN expressiveness analyses characterize aggregation indistinguishability [20, 23]. Our exact declared-feature quotient instead certifies an obstruction for every pointwise scalar scorer on observed occurrences; it is neither graph isomorphism nor a bound on all policies. Structural witnesses also appear in AAMAS universal motion-reachability theory [8], addressing a different guarantee.

LLM algorithm discovery. FunSearch searches priorities inside fixed skeletons [21]; EoH evolves thoughts/code [19]; ReEvo uses reflection [24]. EoH-S optimizes complementary heuristic sets [18]; LACE combines verified interfaces, tools and runtime-constrained portfolio evolution [12]. Offline discovery and cost budgets are

precedents. DRAGON’s per-instance agents transmit boundary constraints and checker feedback [5]. Our distinction is diagnosing whether an interface can express certified conditional rankings and targeting its missing structure. Zero online calls alone establish no advantage.

Abstraction and synthesis. Counterexample-guided refinement uses failure witnesses [6]. Generalized planning learns symbolic features/actions [4], including feature selection and transition separation [10]; DreamCoder learns reusable components with synthesis guidance [9]. We couple aligned resource interventions and feasible boundary replay to sound rankings, equality-join feature selection and repeated full-quotient separation. Acyclicity removes observed information obstructions; bounded-rule fit, shared cost and schedule quality remain distinct requirements.

3 Problem Setting

Each contact $v \in V$ specifies satellite σ_v , ground station γ_v , fixed interval $[s_v, e_v)$ and nonnegative reward w_v . Selection reserves the full interval. Source rewards are duration $e_v - s_v$; synthetic rewards may differ. Configuration θ fixes pairwise compatibility throughout a run (Figure 3).

Graph construction is model-specific. For contacts u, v ordered by start time, a shared ground station creates a conflict when $s_v < e_u + \tau_g$, where $\tau_g \geq 0$ is the station transition time. Shared-satellite conflicts depend on nonnegative change and transition thresholds τ_c, τ_t and the interval relation. Source-derived instances retain their declared satellite predicate, including its admissible crossing overlaps.¹ Synthetic probes use a separately declared single-capacity overlap model. Interventions preserve the declared model.

¹For a common satellite, equal starts or containment create conflicts. A crossing overlap $s_u < s_v < e_u < e_v$ conflicts exactly when $e_u - s_v < \tau_c$; nonoverlapping intervals conflict when $s_v - e_u < \tau_t$.

OFFLINE · evidence-conditioned LLM synthesis with deterministic verification

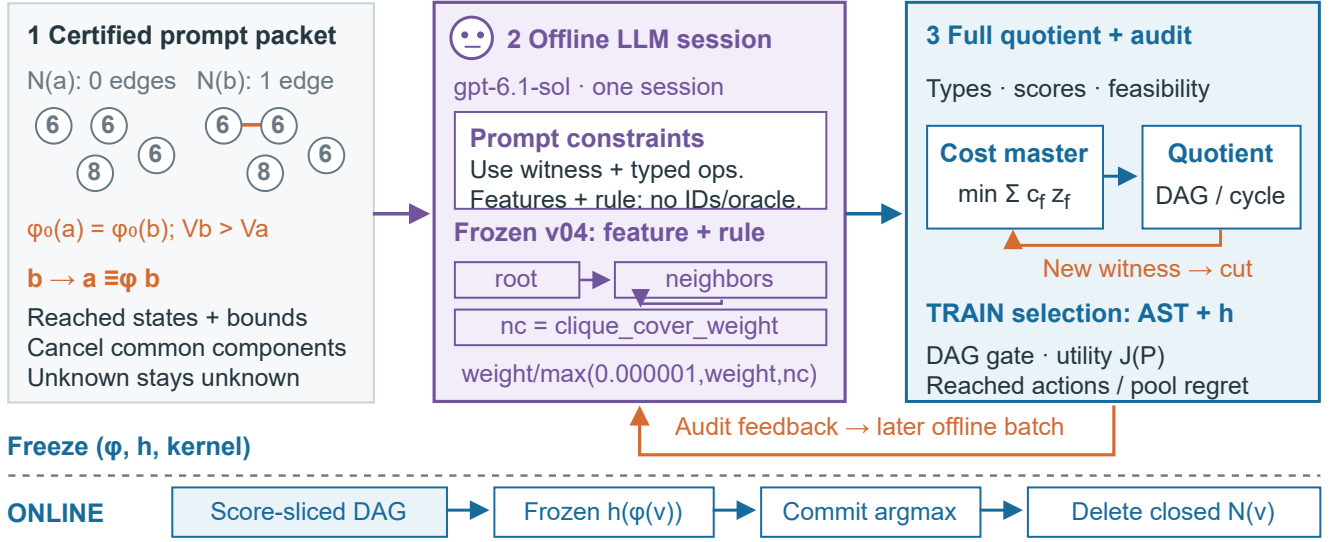


Figure 2: Offline synthesis exposes the LLM’s evidence packet, prompt constraints and joint feature–rule proposals. Deterministic checks repair the full quotient and audit reached actions, quality and compiled cost. Feedback informs later offline batches; the selected score-sliced program is frozen before online greedy commitment and residual updates.

The graph $G_\theta = (V, E_\theta, w)$ joins pairs rejected by these predicates. With $w(I) = \sum_{v \in I} w_v$, contact selection is MWIS:

$$\max_{\substack{I \subseteq V \\ I \text{ independent in } G_\theta}} w(I). \quad (1)$$

Feasibility requires all evaluated hard constraints to be pairwise encoded.

A boundary $B = (F, X)$ records previously selected contacts F and explicit exclusions X . It is admissible when F is independent, $F, X \subseteq V$, and $F \cap X = \emptyset$. Let $N_\theta(F) = \bigcup_{v \in F} N_\theta(v)$, where $N_\theta(v)$ is the neighbor set of v . The available actions are

$$R_\theta(B) = V \setminus (F \cup X \cup N_\theta(F)). \quad (2)$$

For a feasible next action $d \in R_\theta(B)$, define its conditional completion value as

$$V_\theta(d | B) = \max_{\substack{I \text{ independent in } G_\theta \\ F \cup \{d\} \subseteq I, I \cap X = \emptyset}} w(I). \quad (3)$$

This includes fixed reward and globally compatible future contacts.

Aligned configurations θ_0, θ_1 change one resource parameter, sharing contacts, intervals, identities, rewards and an admissible boundary. Competing a, b are available and adjacent on both sides; other changed conflicts can alter their completion values.

A frozen $P = (\phi, h)$ assigns a finite score to every available contact, $h(\phi(G_\theta, r, v))$. The kernel commits an argmax with deterministic ties, deletes its closed neighborhood and repeats until $R = \emptyset$. This guarantees a complete feasible selection. Adaptation uses current graph and permitted resource fields, without online LLM or oracle calls.

4 Certified Representation–Rule Refinement

Offline refinement tests input expressivity, action relevance at reached states, and complete-schedule quality versus measured work (Figure 2). Banks remain fixed during evaluation; the selected feature–rule program is frozen before fresh confirmation and deployment.

4.1 Component-Cancelled Full-Residual Certificates

At an admissible boundary $B = (F, X)$, write $R = R_\theta(B)$ and $R_d = R \setminus (\{d\} \cup N_\theta(d))$. Let C_d be the connected components of $G_\theta[R_d]$, identified by exact vertex sets, and let $\alpha_w(C)$ denote the MWIS value of their induced graph. Additivity gives

$$V_\theta(d | B) = w(F) + w_d + \sum_{C \in C_d} \alpha_w(C). \quad (4)$$

For actions a, b in one graph and boundary, define unmatched collections $\mathcal{A} = C_a \setminus C_b$ and $\mathcal{B} = C_b \setminus C_a$. Shared components have identical weighted induced graphs; their possibly unknown optima cancel:

$$\begin{aligned} \Delta_{ab} &= V_\theta(a | B) - V_\theta(b | B) \\ &= w_a - w_b + \sum_{C \in \mathcal{A}} \alpha_w(C) - \sum_{C \in \mathcal{B}} \alpha_w(C). \end{aligned} \quad (5)$$

Given sound component intervals $[L_C, U_C]$,

$$\begin{aligned} \underline{\Delta}_{ab} &= w_a - w_b + \sum_{\mathcal{A}} L_C - \sum_{\mathcal{B}} U_C, \\ \bar{\Delta}_{ab} &= w_a - w_b + \sum_{\mathcal{A}} U_C - \sum_{\mathcal{B}} L_C. \end{aligned} \quad (6)$$

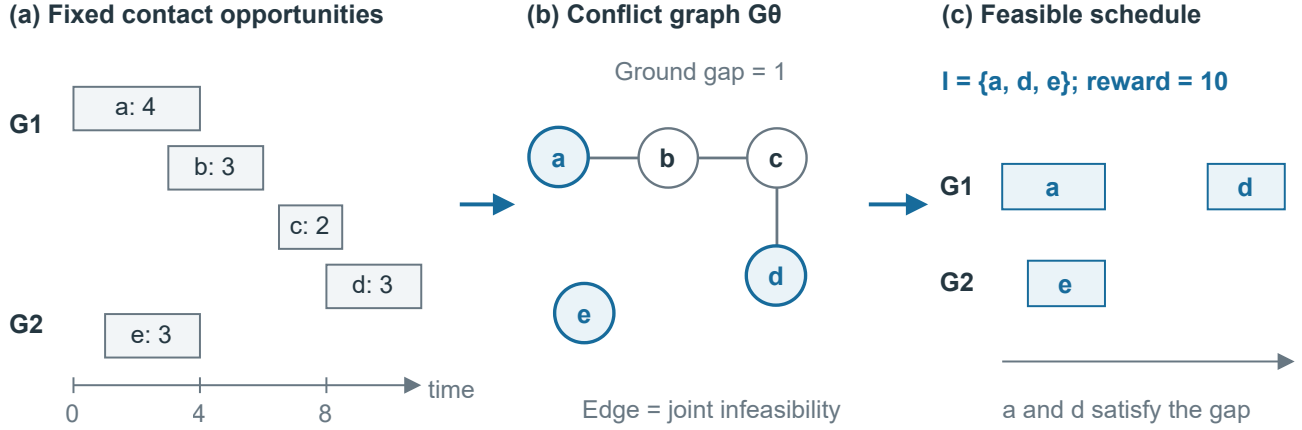


Figure 3: From contact opportunities to a complete feasible schedule: resource predicates create conflict edges; vertex weights encode contact rewards; an independent set selects mutually compatible contact intervals.

Lemma (sound cancellation). The endpoints enclose Δ_{ab} . *Proof.* Independent component optima have a feasible union. Shared terms cancel; bounding unmatched terms with their respective signs proves the enclosure. $\square$

Conditioning can disconnect a graph. Matching uses exact vertex sets within one configuration (Figure 4), not cross-intervention identity.

Checked feasible packings supply lower bounds. For $H \subseteq C$, a sound U_H and verified clique partition $\{K_j\}$ of $C \setminus H$ give

$$U_C = U_H + \sum_j \max_{v \in K_j} w_v. \quad (7)$$

Relaxing cross-region edges proves soundness. Greedy completion and branch-and-bound use exact supplied-weight arithmetic; exported endpoints round outward. Early stopping retains enclosures; large components use zero-search bounds and exhausted calls retain $[0, w(C)]$. Different clique partitions preclude universal interval dominance.

For aligned interventions, replay a, b feasibly at the same B on both sides. With $\epsilon \geq 0$, certify reversal when

$$\underline{\Delta}_{ab}^{\theta_0} > \epsilon, \quad \overline{\Delta}_{ab}^{\theta_1} < -\epsilon, \quad (8)$$

or with roles exchanged. Equal strict signs certify preservation; unresolved intervals and ties remain unknown at budget exhaustion. Certified sides supply occurrences $x = (G_\theta, R, v)$ and requirements

$$h(\phi(x^+)) > h(\phi(x^-)). \quad (9)$$

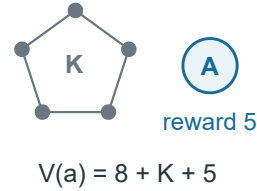
Objective tolerance ϵ differs from a scoring margin.

4.2 Exact Quotients and Finite-Catalogue Repair

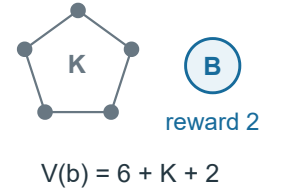
For finite evidence, Q_ϕ identifies occurrences exactly when $\phi(x) = \phi(y)$ and adds arcs $[x^+] \rightarrow [x^-]$, retaining self-loops. The nine base inputs are weight, duration, residual degree, neighbor-weight sum/maximum, compatible-weight sum, ground/satellite gaps and residual count. Diagnosis includes every declared field, even if ignored by the rule; IDs/split labels are excluded. A configuration

Cancel identical residual components

(a) Commit a: reward 8



(b) Commit b: reward 6



Same vertex set K ; $\text{MWIS}(K) \in [20, 30]$

Separate intervals overlap

$V(a)$: [33, 43]

$V(b)$: [28, 38]

$\Delta = (8 + 5) - (6 + 2) = 5 \rightarrow \text{prefer } a$
Zero search nodes; common K never queried

Figure 4: Identical residual components cancel from the global conditional-value difference. In this constructed zero-search example, separate intervals [33, 43] and [28, 38] overlap, yet shared five-cycle K cancels exactly, leaving $\Delta = 5$ and a strict preference for a . The oracle never bounds K .

audit adds omitted satellite transition time and presence before attributing aliases to graph structure.

Proposition 1 (finite representability). An unrestricted deterministic pointwise scalar score satisfies every observed strict requirement if and only if Q_ϕ is acyclic. *Proof.* Cycles demand a strict decrease back to the same score. Longest outgoing-path lengths on a finite DAG satisfy every arc and scale to any positive margin. $\square$

A concrete cycle records ranking arcs $p_i \rightarrow n_i$ and equality joins $n_i \sim_\phi p_{i+1}$: distinctions that LLM features should recover. For finite typed catalogue $\mathcal{F}$, let $d_{Wf} = 1$ when f separates any join of witness W . With positive standalone costs c_f , solve

$$\begin{aligned} \min_{z \in \{0,1\}^{|\mathcal{F}|}} \quad & \sum_f c_f z_f, \\ \text{s.t.} \quad & \sum_f d_{Wf} z_f \geq 1 \quad \text{for each retained } W, \\ & \sum_f z_f \leq K. \end{aligned} \quad (10)$$

Trials replace previous additions to the fixed base and rebuild the full quotient. Surviving cycles add necessary cuts: splitting a coarse cycle can leave refined cycles projecting to repeated-class closed walks.

Proposition 2 (finite-catalogue repair). Globally solved masters and a final acyclic full quotient minimize additive cost over fixed catalogue, evidence and K . *Proof.* Necessary witness constraints make the master optimum a lower bound on valid repair cost. The final DAG attains that bound. $\square$

With variable K , arbitrary finite tables encode weighted set cover: one self-loop pair per element, separated by covering-subset features. Hardness need not hold for every grammar; fixed K permits $O(|\mathcal{F}|^K)$ trials. Exhaustion/uncovered witnesses remain unresolved. Acyclicity does not guarantee bounded-rule fit or transfer. Minimum-interface search remains an ablation.

4.3 Auditing Actual Reached Actions

Requirements can concern unchosen actions. Rollouts record the initial boundary and up to two commitments. Audits retain the initial and most-shared boundary, deterministic ties and unsampled-state counts.

Replay each program’s deterministic argmax at the same B , forming finite pool $\mathcal{U}_B$. Query all unordered pairs in fixed order under shared cache/budget. For chosen d , define

$$\rho_{\mathcal{U}_B}(d) = \max_{u \in \mathcal{U}_B} [V_\theta(u \mid B) - V_\theta(d \mid B)]. \quad (11)$$

Since $d \in \mathcal{U}_B$, maxima of zero and difference endpoints enclose ρ . The upper endpoint bounds only finite-pool regret; the lower can prove loss to a particular feasible alternative. Neither guarantees complete-schedule loss. Counterfactual states remain auditable, but selection uses only boundaries in that program’s recorded rollout. Coverage, failed choices and unknown comparisons accompany regret; strict differences augment quotient evidence.

4.4 Typed Features and Demand-Driven Execution

The language types nodes, node/edge sets and numbers; proposals pass type, size and finite-output checks. Set operations, induced edges and bounded arithmetic compose packing/cover and edge statistics, including summaries of R_v . For example,

$$T_R(v) = \sum_{\{u,z\} \in E_\theta[N_R(v)]} \min(w_u, w_z) \quad (12)$$

repeatedly counts shared endpoints, making it a statistic, not a completion bound. Rounded packing/cover scorer inputs are heuristic; the independent sound offline bound oracle uses exact arithmetic to certify labels (Figure 5).

State-wide guards $|R| \leq 64$ or 128 choose $w_v + (L_v + U_v)/2$ using rounded packing/cover summaries of R_v , otherwise density/pressure scoring. Every action shares the branch; midpoints give no ranking guarantee.

Shared DAGs and lazy conditional/short-circuit inputs avoid untaken generic primitives; static aggregate-superset initialization/maintenance stays charged. For defined finite eager features, scores, ties and complete traces are preserved. Diagnosis retains the full interface. Sequential deletion uses

$$T(v) \leftarrow T(v) - \sum_{y \in N_R(v) \cap N_R(x)} \min(w_x, w_y). \quad (13)$$

Updated neighborhoods prevent double subtraction; exact accumulators preserve rounded reference terms. Unsupported expressions use the snapshot interpreter. All compilation, cache, update and scoring work is charged; joint DAG cost is nonadditive, unlike c_f .

A separate backend caches only reviewed root/neighborhood dependencies under immutable graph, weights, configuration and program. After deleting the whole batch D , it refreshes survivors in the pre-deletion frontier $\bigcup_{x \in D} N_R(x)$; outside this frontier, unchanged inputs preserve scores. Versioned heap entries preserve scalar-score/string-ID ordering. Global or unreviewed demanded inputs, even in untaken branches, trigger full rescoring. Initialization, frontier refreshes, comparisons and stale removal are charged. For valid queried inputs this preserves the score-sliced kernel’s trace and frozen feature–rule pair.

4.5 TRAIN Selection and Frozen Deployment

One offline assistant session authors proposals before TRAIN outcomes. Controls have unequal authoring counts/compute. Numerical selection uses TRAIN only, excluding fresh validation/test outcomes.

For context t , qp_t divides reward by the best completed bank reward, a feasible empirical comparator; cp_t divides measured work by positive degree-baseline work. Incomplete runs receive $qp_t = 0$, $cp_t = 100$. Equal-family means define $q(P)$, $c(P)$ and

$$J(P) = q(P) - 0.002[c(P) - 1]. \quad (14)$$

New guided, free and enumerated arms require acyclic interface quotients; other controls retain separate scope. Programs within 0.002 of each arm’s highest utility rank by reached pool-regret upper endpoint, then lower endpoint, unknown count, utility and fixed order. Regret uses full-context normalization $\max(1, \text{mean}_{v \in V} w_v)$ and equal-family means. Missing coverage sorts last; unknowns are not exclusion gates. Primary-only guided selection remains an ablation.

All programs use the common demanded backend and cooperative five process-CPU-second target; overshoot/completion are recorded. Failure returns no schedule or fallback. Bank, evidence, program, compiler and configuration freeze before confirmation. The kernel guarantees feasibility; quality/transfer remain empirical.

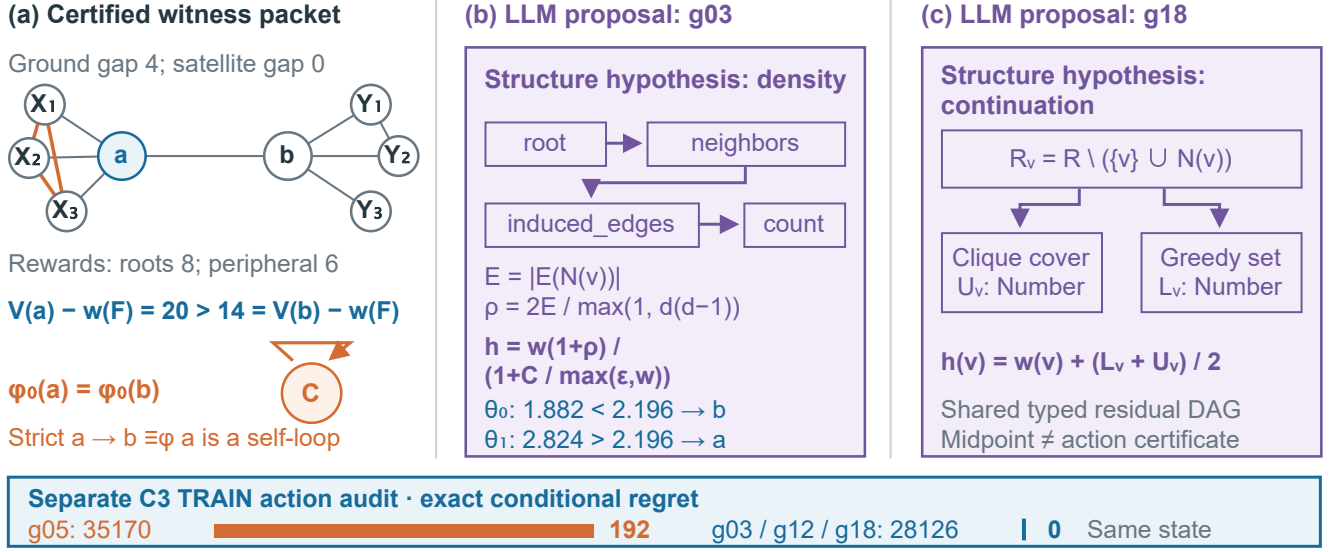


Figure 5: A constructed diagnostic self-loop exposes missing neighborhood structure. The actual assistant-authored g03 density and g18 continuation programs use distinct typed ASTs and ranking hypotheses. g03 separates both probe preferences; the separate TRAIN action audit illustrates why witness separation must still be followed by full-quotient, action and cost checks.

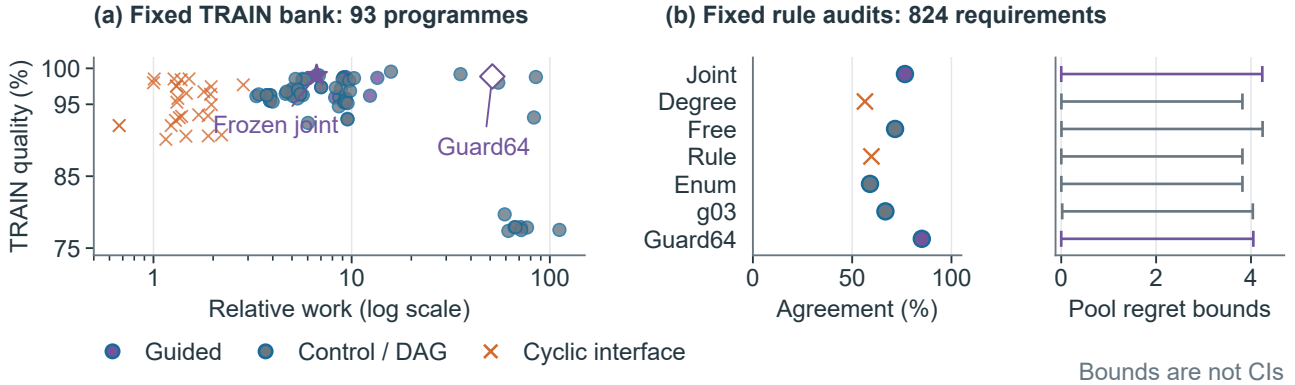


Figure 6: TRAIN evidence separates input representability, rule fit and computation. Acyclic interfaces need not fit every requirement: the frozen primary satisfies 631/824. Guard64 satisfies 701/824 but costs more. Rightmost intervals are sound finite-pool regret bounds, not confidence intervals.

5 Experiments

5.1 Populations and Frozen Protocol

TRAIN has 33 pairs/66 contexts at 32–128 contacts. Fresh confirmation uses 456 test contexts; 132 validation contexts do not select programs (Table 1). Pairs preserve contacts, weights and windows while changing one resource parameter. Three resource regimes share seeds. Test ground gaps (0.5, 6) shift from TRAIN (0, 2); dense horizons shorten $1.8n \rightarrow 0.18n$ and maximum duration grows $12 \rightarrow 48$.

Seeds/C3 opportunities are split-disjoint and exclude inventoried prior inputs. Public transfer uses 18 DIMACS originals and 30 SATLIB instances with unit/hash weights; independent decoders

Table 1: Separate confirmation populations. Weight variants share topology; scheduling configurations share paired sources.

Population	Contacts	Contexts
Fresh standard / dense-long	64–256	216 / 216
Fresh C3 source blocks	64–512	24
DIMACS / SATLIB transfer	28–1,024	36 / 60
Separate SNAP sparse extension	4,039–12,008	8

check clique complements and literal-conflict reductions. SNAP projects four sources to simple graphs.

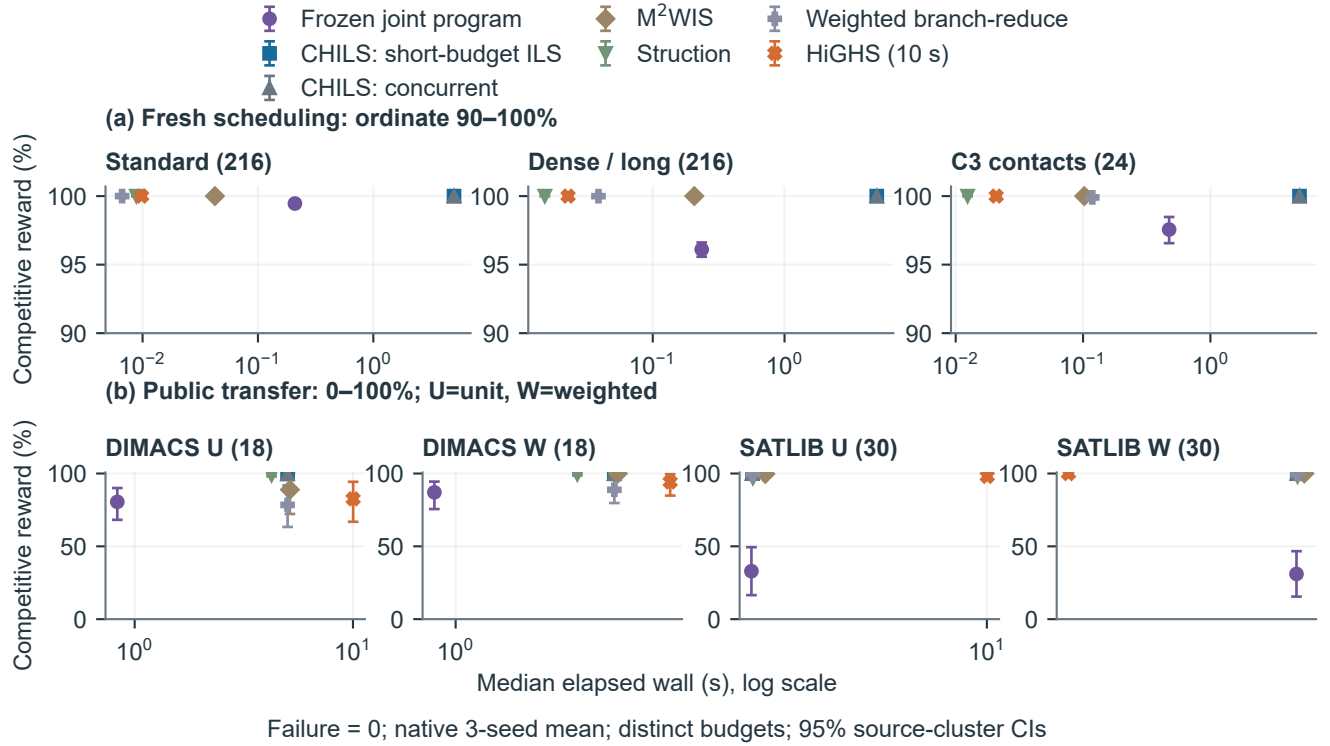


Figure 7: Complete-assignment quality and cost against published methods. Competitive reward uses the best saved verified feasible witness; failures receive zero. Native quality averages three seeds. Bootstrap intervals cluster source instances. The two rows use different ordinate ranges. Budgets differ; public failures and native superiority are retained.

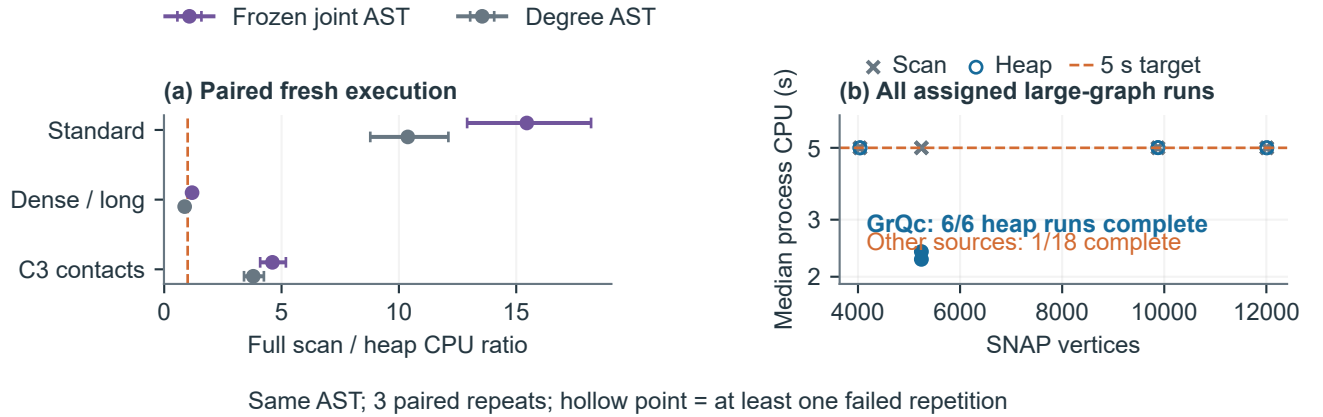


Figure 8: Execution adaptation preserves the frozen score rule. All 2,736 assessable fresh pairs have identical traces; ratios use context medians of three repeats with source-cluster intervals. Sparse full scans never complete within the five-second CPU target, so their speedups and trace parity remain unassessable.

Pinned CHILS, M²WIS, Struction and weighted branch-and-reduce executables [11, 13–15] have verified output mappings and seeds 1/2/3. CHILS uses four population states and a one-state ILS control. Native budgets are five seconds or 30 on prespecified

subsets. Constructive policies retain a cooperative five-process-CPU-second target; HiGHS receives ten seconds and classical one-to-two exchange search two after initialization. Computation is unmatched.

Table 2: Attempts, reversal/preservation certificates and expanded nodes on the same 62 pairs.

Challenger / envelope	Attempts	R/P	Nodes
Ranked / clique	172	26/25	52,355
Uniform / clique	170	14/44	52,377
Ranked / weight sum	176	24/12	84,138

Null failures receive zero all-assigned reward; coverage accompanies conditional summaries. Competitive reward averages native seeds within contexts and divides by the strongest verified saved feasible witness, without establishing optimality. Formal clique-upper ratios remain separate from floating HiGHS bounds. The 2,000-replicate 95% bootstrap retains paired configurations/shared seeds across regimes and C3 source clusters; public family/weight modes remain separate. Wall points average native seeds; standalone exchange-search cost includes initialization.

5.2 Evidence and Cancellation

Acquisition uses 62 training pairs: 20 probes and 42 temporal/C3 pairs. Matched attempt limits/oracle parameters compare challengers and envelopes (Table 2). Ranked clique improves on weight sums with less search but yields fewer certificates than uniform. Extra reversals mainly arise from probes; repeated boundary certificates are clustered.

All 64 original strict self-loops come from probes; 19 non-probe specifications form an acyclic quotient and 200 broader alias queries remain unresolved. Catalogue repair proves neither natural incidence nor unrestricted minimality; full-quotient checks detect cycles surviving coarse-witness repair.

Pure TRAIN cancellation fixes unmatched bounds on 396 comparisons. All intervals remain nested; 199 narrow, with median width $86.75 \rightarrow 36$ reward units. Strict certificates remain $33 \rightarrow 33$. The separate whole-residual/component $10 \rightarrow 33$ comparison changes strategy and does not isolate cancellation.

5.3 Frozen Selection and Confirmation

The 93-program bank contains 12 new guided proposals, 24 each from existing free/rule-only/enumeration arms, five classical rules and four prior guided controls. One Codex session (gpt-6.1-sol, ultra) authors new proposals without external API calls. Unequal counts and unavailable tokens preclude model-level or matched-compute guidance claims.

TRAIN completes 5,996/6,138 assignments. The selected rule is

$$h(v) = \frac{w_v}{\max\{10^{-6}, w_v, nc_v\}}.$$

Here nc is the deterministic weighted clique-cover envelope of active neighbors. Both guided selections coincide; the secondary criterion has no observed effect.

Retrospective six-program auditing covers 588 contexts. Across 555 sampled primary reached test states, finite-pool regret is positive in 18, zero in 191 and unresolved in 346; counterfactual states do not enter these counts.

Primary completion is 456/456 fresh contexts. Competitive reward is 99.46% [99.36,99.55] on standard, 96.11% [95.58,96.62] on

dense and 97.56% [96.57,98.48] C3 (Figure 7). Degree-rule advantages are 0.98 percentage points [0.79,1.17] and 1.64 [0.93,2.36] temporally; the C3 interval crosses zero. CHILS, ILS, M²WIS and Struction attain 100% descriptive competitive reward. M²WIS, Struction and weighted branch-and-reduce terminate faster; no published-solver superiority follows. Older continuation fails 211/456. Completed C3 schedules pass the original physical verifier; only two fresh contexts attain the clique upper.

Public primary completion is 34/36 DIMACS and 20/60 SATLIB. Unit-weight competitive rewards are 80.53% and 32.93%; hash-weight rewards are 87.07% and 31.05%, respectively (Figure 7). CHILS, ILS and Struction complete 288/288 short seed runs; M²WIS and weighted branch-and-reduce fail six/eight. All four native methods complete 63/63 subset runs; primary completion is 18/21 at unchanged CPU target. SNAP primary completion is 0/8 versus 24/24 native seeds each except weighted branch-and-reduce, 21/24.

5.4 Execution Cost

Demanded/full-interface primary traces match on 456 fresh contexts. A separate post-diagnostic heap extension preserves frozen ASTs and all 2,736 assessable fresh pairs (Figure 8). Mean context-median full-scan/heap CPU ratios are $15.45 \times [12.91, 18.19]$, $1.19 \times [1.15, 1.23]$ and $4.61 \times [4.09, 5.19]$ on standard/dense/C3. Degree slows on dense graphs (0.87 $\times$). SNAP full scans complete 0/24 repeats per AST; heap completes 7/24 primary and 18/24 degree runs. Sparse parity/speedups remain unassessable; failed-run quality receives no credit in original confirmation.

6 Discussion and Limits

Contradictions are certified on designed probes; all 200 broader natural alias queries remain unresolved. Preference change alone is insufficient. The primary interface is acyclic, yet its rule satisfies only 631/824 TRAIN requirements (788 actual-action, 36 prior). The DAG gate implies unrestricted finite representability, not bounded-rule fit or unseen consistency.

Repair optimality covers fixed evidence, a bounded finite catalogue and additive standalone costs. Shared cost and schedule quality remain empirical. Finite-pool regret excludes other feasible actions; unresolved bounds remain unknown. Controlled cancellation narrows uncertainty without strict-yield gain.

Unequal proposal counts/checking effort and one authoring session preclude model-level guidance claims. Fresh seeds/contacts prevent instance reuse, but C3 remains one physical source. Public hash weights/simple projections are declared extensions. Feasibility covers encoded predicates; results establish no global greedy optimality or published-solver dominance.

7 Conclusion

Certified conditional preferences expose lost distinctions through exact quotient contradictions. Equality-join witnesses direct typed feature synthesis; full-quotient separation selects finite-catalogue repairs. Complete schedules and compiled work guide feature-rule selection. The frozen program adapts within a feasibility-enforcing kernel without online LLM or oracle calls, making information adequacy explicit while separating its guarantees from measured performance.

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